

Krishna Soni

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Summary

AI/ML engineer specializing in Retrieval-Augmented Generation and agentic systems, with hands-on experience building production-grade RAG pipelines, multi-agent research frameworks, and LangGraph-based recruiting workflows. Focused on applying evaluation-driven development to move beyond prototype-stage AI systems toward measurable, deployable products.

Education

University of Petroleum and Energy Studies, Bachelor of Technology in ECE
CGPA: 7.85

August 2025 – May 2029

Skills

Languages and Frameworks: Python (FastAPI), C++, JavaScript, TypeScript (Node.js).

Databases and Caching: PostgreSQL, MySQL, MongoDB, Redis, ChromaDB, Qdrant (Vector DBs).

Infrastructure & DevOps: Docker, AWS (EC2/S3), Git, CI/CD, MCP (Model Context Protocol), Streamlit Cloud.

Concepts: Retrieval-Augmented Generation (RAG), Hybrid Search, Agentic Workflows, Neural Networks, Supervised/Unsupervised Learning, Fine-Tuning, Data Pipelines, Feature Engineering, Evaluation Metrics (recall@k, MRR, nDCG).

AI/ML Frameworks: PyTorch, TensorFlow, LangChain (AI Agents), LangGraph, Scikit-learn, Hugging Face, OpenCV, Deep Learning, NLP.

Projects

AI Research Framework Live link

[GitHub Repo](#)

- Designed a cooperative network of autonomous **AI agents** (`agents.py`) with distinct roles and independent LLM configurations to parallelize research tasks.
- Built a stateful task pipeline (`pipeline.py`) separating web discovery, content synthesis, and report generation; exposed agent tools via a decorator-based system (`tool.py`) with built-in error handling.
- Shipped an interactive **Streamlit** dashboard with deterministic management using the **uv** package engine.

AI Video Assistant with RAG Pipeline

[GitHub Repo](#)

- Engineered a multimodal pipeline that ingests raw video, extracts audio, and generates clean transcripts for downstream querying.
- Implemented a **RAG** workflow using **Mistral AI** with transcript chunking for precise, context-aware answers without overloading the LLM's context window.

AI Recruiting Matching Agent

[GitHub Repo](#)

- Built an agentic **LangGraph** pipeline with a multi-node workflow (**JD parsing**, **requirement extraction**, **RAG-based resume search**, **multi-round candidate ranking**) and a human-in-the-loop feedback node.
- Implemented a **RAG search** tool using **Mistral embeddings** and **ChromaDB** for semantic matching.
- Built an evaluation harness measuring retrieval quality against hand-labeled ground truth, achieving **0.92 recall@3** and **0.90 nDCG@3** across 6 test queries spanning in-domain and out-of-domain resumes.

Certifications

- Supervised Machine Learning: Regression and Classification** – Issued by DeepLearning.AI & Stanford University on Coursera [[View Certificate](#)]
- Software Engineer Intern Certification** – Verified by HackerRank [[View Certificate](#)]
- Advanced Learning Algorithms** – [[View Certificate](#)]
- Data Science Job Simulation Certificate** – Lloyds Banking Group's Data Science on Forage [[View Certificate](#)]